Math 20
Module 1: Equations and Inequalities

Nile Jocson
<atraphaxia@gmail.com>

February 19, 2025

Contents

1 **5**
1.1 . . . 5
1.2 . . . 6
1.3 . . . 7
1.4 . . . 8
1.5 . . . 9
1.6 . . . 10
1.7 . . . 11
1.8 . . . 12
1.9 . . . 13
1.10 . . . 14
1.11 . . . 15
1.12 . . . 16
1.13 . . . 17
1.14 . . . 18
1.15 . . . 19
1.16 . . . 20
1.17 . . . 21
1.18 . . . 22
1.19 . . . 23
1.20 . . . 24
1.21 . . . 25
1.22 . . . 26
1.23 . . . 27
1.24 . . . 28
1.25 . . . 29
1.26 . . . 30
1.27 . . . 31
1.28 . . . 32
1.29 . . . 33
1.30 . . . 34
1.31 . . . 35
1.32 . . . 36
1.33 . . . 37
1.34 . . . 38
1.35 . . . 40
1.36 . . . 41
1.37 . . . 42
1.38 . . . 43
1.39 . . . 44
1.40 . . . 45
1.41 . . . 46
1.42 . . . 47
1.43 . . . 48
1.44 . . . 49
1.45 . . . 50
1.46 . . . 51
1.47 . . . 52
1.48 . . . 53
1.49 . . . 54
1.50 . . . 55
1.51 . . . 56
1.52 . . . 57
1.53 . . . 58
1.54 . . . 59
1.55 . . . 60
1.56 . . . 61
1.57 . . . 62
1.58 . . . 63

1.59 64
1.60 65
1.61 66
1.62 67
1.63 68
1.64 69
1.65 70
1.66 71

2 **72**
2.1 72

3 **73**
3.1 73
3.2 74
3.3 75
3.4 76
3.5 77
3.6 78
3.7 79
3.8 80
3.9 81
3.10 82
3.11 83
3.12 84
3.13 85
3.14 86
3.15 87
3.16 88
3.17 89
3.18 90
3.19 91
3.20 92
3.21 93
3.22 94
3.23 95
3.24 96
3.25 97
3.26 98
3.27 99
3.28 100
3.29 101
3.30 102
3.31 103
3.32 104
3.33 105
3.34 106
3.35 107
3.36 108
3.37 109
3.38 110
3.39 111
3.40 112
3.41 113
3.42 114
3.43 115
3.44 116

4 **117**
4.1 117
4.2 118
4.3 119
4.4 120
4.5 121
4.6 122
4.7 123
4.8 124
4.9 125
4.10 126
4.11 127
4.12 128
4.13 129
4.14 130
4.15 131
4.16 132
4.17 133

1

Solve the following equations.

1.1

$$x + 9 = 5 - 3x$$

$x + 3x = 5 - 9$ $4x = -4$	Isolate x.
$x = -1$	Final answer. ■

1.2

$$\frac{2x+3}{4} - \frac{x+1}{2} = -\frac{1}{3}$$

$12(\frac{2x+3}{4} - \frac{x+1}{2}) = 12(-\frac{1}{3})$	lcm $= 12$
$3(2x+3) - 6(x+1) = -4$	
$\cancel{6x} + 9 - \cancel{6x} - 6 = -4$	
$9 - 6 = -4$	
$3 = -4$	
$x \in \emptyset$	Final answer.
	$\blacksquare$

1.3

$$3x^2 - 2x + 1 = 0$$

$x = \frac{-(-2)\pm\sqrt{(-2)^2-4(3)(1)}}{2(3)}$ $x = \frac{2\pm\sqrt{4-12}}{6}$ $x = \frac{2\pm\sqrt{-8}}{6}$ $x = \frac{2\pm i\sqrt{4}\sqrt{2}}{6}$ $x = \frac{2\pm 2i\sqrt{2}}{6}$	Use the quadratic formula.
$x = \frac{1\pm i\sqrt{2}}{3}$ ■	Final answer.

1.4

$$4x^2 + 2x = 2$$

$4x^2 + 2x - 2 = 0$	Rewrite in standard form.
$2(2x^2 + x - 1) = 0$	
$(2x - 1)(x + 1) = 0$	Factor by grouping.
$x \in \{\frac{1}{2}, -1\}$	Final answer.
	■

1.5

$$16x^2 + 9 = 24x$$

$16x^2 - 24x + 9 = 0$	Rewrite in standard form.
$16x^2 - 12x - 12x + 9 = 0$	Factor by grouping.
$(4x - 3)^2 = 0$	
$x = \frac{3}{4}$	Final answer.
	■

1.6

$$\frac{x}{x-1} + \frac{x-5}{x^2+2x-3} = \frac{1}{x+3}$$

	$\frac{x}{x-1} + \frac{x-5}{(x-1)(x+3)} = \frac{1}{x+3}$	Factor by grouping.
	$(x-1)(x+3)(\frac{x}{x-1} + \frac{x-5}{(x-1)(x+3)}) = (x-1)(x+3)(\frac{1}{x+3})$	lcm $= (x-1)(x+3)$
$x \notin \{1, -3\}$	$x(x+3) + x - 5 = x - 1$	
$x \notin \{1, -3\}$	$x^2 + 3x + \cancel{x} - 5 = \cancel{x} - 1$	
$x \notin \{1, -3\}$	$x^2 + 3x - 4 = 0$	
$x \notin \{1, -3\}$	$(x+4)(x-1) = 0$	Factor by grouping.
$x \notin \{1, -3\}$	$x \in \{-4, 1\}$	
	$x = -4$	Final answer.
		■

1.7

$$\frac{1}{4-x} + \frac{3}{6+x} = 0$$

	$(4-x)(6+x)(\frac{1}{4-x} + \frac{3}{6+x}) = 0$	lcm $= (4-x)(6+x)$
$x \notin \{4, -6\}$	$6 + x + 3(4-x) = 0$	
$x \notin \{4, -6\}$	$6 + x + 12 - 3x = 0$	
$x \notin \{4, -6\}$	$x - 3x = -6 - 12$	
$x \notin \{4, -6\}$	$2x = -18$	
	$x = 9$	Final answer.
		■

1.8

$$\frac{3}{x^2-9} - \frac{7}{x-3} = -\frac{4}{x+3}$$

	$\frac{3}{(x-3)(x+3)} - \frac{7}{x-3} = -\frac{4}{x+3}$	Factor using difference of two squares.
	$(x-3)(x+3)(\frac{3}{(x-3)(x+3)} - \frac{7}{x-3}) = (x-3)(x+3)(-\frac{4}{x+3})$	$lcm = (x-3)(x+3)$
$x \notin \{3, -3\}$	$3 - 7(x+3) = -4(x-3)$	
$x \notin \{3, -3\}$	$3 - 7x - 21 = -4x + 12$	
$x \notin \{3, -3\}$	$-7x + 4x = 12 + 21 - 3$	
$x \notin \{3, -3\}$	$-3x = 30$	
	$x = -10$	Final answer.
		■

1.9

$$\frac{3}{x^2 - x - 6} = \frac{4}{2x^2 + x - 6}$$

	$\frac{3}{(x-3)(x+2)} = \frac{4}{(2x-3)(x+2)}$	Factor by grouping.
	$(x-3)(2x-3)(x+2)(\frac{3}{(x-3)(x+2)}) = (x-3)(2x-3)(x+2)(\frac{4}{(2x-3)(x+2)})$	$lcm = (x-3)(2x-3)(x+2)$
$x \notin \{3, \frac{3}{2}, -2\}$	$3(2x-3) = 4(x-3)$	
$x \notin \{3, \frac{3}{2}, -2\}$	$6x - 9 = 4x - 12$	
$x \notin \{3, \frac{3}{2}, -2\}$	$6x - 4x = -12 + 9$	
$x \notin \{3, \frac{3}{2}, -2\}$	$2x = -3$	
	$x = -\frac{3}{2}$	Final answer.
		■

1.10

$$\frac{2}{y+1} - \frac{3}{1-y} = \frac{5}{y}$$

	$(y+1)(1-y)(\frac{2}{y+1} - \frac{3}{1-y}) = y(y+1)(1-y)(\frac{5}{y})$	lcm $= y(y+1)(1-y)$
$y \notin \{0, -1, 1\}$	$2y(1-y) - 3y(y+1) = 5(y+1)(1-y)$	
$y \notin \{0, -1, 1\}$	$2y - 2y^2 - 3y^2 - 3y = 5(1-y^2)$	$y+1 = 1+y$, expand using difference of two squares.
$y \notin \{0, -1, 1\}$	$-2y^2 - 3y^2 - 3y + 2y = 5 - 5y^2$	
$y \notin \{0, -1, 1\}$	$\cancel{-5y^2} - y = 5 - \cancel{5y^2}$	
	$y = -5$	Final answer.
		■

1.11

$$\frac{x+17}{x^2-6x+8}+\frac{x-2}{x-4}=\frac{x-4}{x-2}$$

	$\frac{x+17}{(x-2)(x-4)}+\frac{x-2}{x-4}=\frac{x-4}{x-2}$	Factor by grouping.
	$(x-2)(x-4)(\frac{x+17}{(x-2)(x-4)}+\frac{x-2}{x-4})=(x-2)(x-4)(\frac{x-4}{x-2})$	lcm $=(x-2)(x-4)$
$x\notin\{2,4\}$	$x+17+(x-2)^2=(x-4)^2$	
$x\notin\{2,4\}$	$x+17+x^2-4x+4=x^2-8x+16$	
$x\notin\{2,4\}$	$\cancel{x^2}-4x+x+4+17=\cancel{x^2}-8x+16$	
$x\notin\{2,4\}$	$-3x+21=-8x+16$	
$x\notin\{2,4\}$	$-3x+8x=16-21$	
$x\notin\{2,4\}$	$5x=-5$	
	$x=-1$	Final answer.
		■

1.12

$$\frac{w}{3w^2 - 8w + 4} = \frac{w+2}{3w^2 + w - 2}$$

	$\frac{w}{3w^2-6w-2w+4} = \frac{w+2}{3w^2+3w-2w-2}$	Factor by grouping.
	$\frac{w}{(3w-2)(w-2)} = \frac{w+2}{(3w-2)(w+1)}$	
	$(3w-2)(w-2)(w+1)(\frac{w}{(3w-2)(w-2)}) = (3w-2)(w-2)(w+1)(\frac{w+2}{(3w-2)(w+1)})$	lcm $= (3w-2)(w-2)(w+1)$
$w \notin \{\frac{2}{3}, 2, -1\}$	$w(w+1) = (w+2)(w-2)$	
$w \notin \{\frac{2}{3}, 2, -1\}$	$\cancel{w^2} + w = \cancel{w^2} - 4$	Expand using difference of two squares.
	$w = -4$	Final answer.

■

1.13

$$\frac{3}{x+4} = \frac{2}{3x-2}$$

	$(x+4)(3x-2)(\frac{3}{x+4}) = (x+4)(3x-2)(\frac{2}{3x-2})$	lcm $= (x+4)(3x-2)$
$x \notin \{-4, \frac{2}{3}\}$	$3(3x-2) = 2(x+4)$	
$x \notin \{-4, \frac{2}{3}\}$	$9x - 6 = 2x + 8$	
$x \notin \{-4, \frac{2}{3}\}$	$9x - 2x = 8 + 6$	
$x \notin \{-4, \frac{2}{3}\}$	$7x = 14$	
	$x = 2$	Final answer.
		■

1.14

$$\frac{x+4}{x+2} = \frac{2x-3}{x-5}$$

	$(x+2)(x-5)(\frac{x+4}{x+2}) = (x+2)(x-5)(\frac{2x-3}{x-5})$	lcm $= (x+2)(x-5)$
$x \notin \{-2, 5\}$	$(x+4)(x-5) = (2x-3)(x+2)$	
$x \notin \{-2, 5\}$	$x^2 - x - 20 = 2x^2 + x - 6$	
$x \notin \{-2, 5\}$	$x^2 - 2x^2 - x - x - 20 + 6 = 0$	Rewrite in standard form.
$x \notin \{-2, 5\}$	$-x^2 - 2x - 14 = 0$	
$x \notin \{-2, 5\}$	$x^2 + 2x + 14 = 0$	
$x \notin \{-2, 5\}$	$x = \frac{-2 \pm \sqrt{2^2 - 4(1)(14)}}{2(1)}$	Use the quadratic formula.
$x \notin \{-2, 5\}$	$x = \frac{-2 \pm \sqrt{4-56}}{2}$	
$x \notin \{-2, 5\}$	$x = \frac{-2 \pm \sqrt{-52}}{2}$	
$x \notin \{-2, 5\}$	$x = \frac{-2 \pm i\sqrt{4}\sqrt{13}}{2}$	
$x \notin \{-2, 5\}$	$x = \frac{-2 \pm 2i\sqrt{13}}{2}$	
	$-1 \pm i\sqrt{13}$	Final answer.

■

1.15

$$\frac{4}{z-2} = \frac{z}{6}$$

	$6(z-2)(\frac{4}{z-2}) = 6(z-2)(\frac{z}{6})$	lcm $= 6(z-2)$
$z \neq 2$	$6(4) = z(z-2)$	
$z \neq 2$	$24 = z^2 - 2z$	
$z \neq 2$	$z^2 - 2z - 24 = 0$	Rewrite in standard form.
$z \neq 2$	$(z-6)(z+4) = 0$	
	$x \in \{6, -4\}$	Final answer.
		■

1.16

$$\frac{1}{2x+5} = \frac{4}{2x-1}$$

	$(2x+5)(2x-1)(\frac{1}{2x+5}) = (2x+5)(2x-1)(\frac{4}{2x-1})$	lcm $= (2x+5)(2x-1)$
$x \notin \{-\frac{5}{2}, \frac{1}{2}\}$	$2x-1 = 4(2x+5)$	
$x \notin \{-\frac{5}{2}, \frac{1}{2}\}$	$2x-1 = 8x+20$	
$x \notin \{-\frac{5}{2}, \frac{1}{2}\}$	$2x-8x = 20+1$	
$x \notin \{-\frac{5}{2}, \frac{1}{2}\}$	$-6x = 21$	
	$x = -\frac{7}{2}$	Final answer.
		■

1.17

$$\frac{2}{3x} + \frac{4}{5x} = \frac{11}{15}$$

	$15x(\frac{2}{3x} + \frac{4}{5x}) = 15x(\frac{11}{15})$	lcm $= 15x$
$x \neq 0$	$10 + 12 = 11x$	
$x \neq 0$	$22 = 11x$	
	$x = 2$	Final answer.
		■

1.18

$$\frac{5}{x+3} = \frac{x+8}{x+3} - 3$$

$x \neq -3$	$5 = x + 8 - 3(x+3)$	Eliminate denominator.
$x \neq -3$	$5 = x + 8 - 3x - 9$	
$x \neq -3$	$5 = -2x - 1$	
$x \neq -3$	$2x = -1 - 5$	
$x \neq -3$	$2x = -6$	
$x \neq -3$	$x = -3$	
	$x \in \emptyset$	Final answer.
		■

1.19

$$\frac{3x}{x+2} = \frac{x+4}{x}$$

	$x(x+2)(\frac{3x}{x+2}) = x(x+2)(\frac{x+4}{x})$	lcm $= x(x+2)$
$x \notin \{0, -2\}$	$3x^2 = (x+2)(x+4)$	
$x \notin \{0, -2\}$	$3x^2 = x^2 + 6x + 8$	
$x \notin \{0, -2\}$	$3x^2 - x^2 - 6x - 8 = 0$	Rewrite in standard form.
$x \notin \{0, -2\}$	$2x^2 - 6x - 8 = 0$	
$x \notin \{0, -2\}$	$x^2 - 3x - 4 = 0$	
$x \notin \{0, -2\}$	$(x-4)(x+1) = 0$	Factor by grouping.
	$x \in \{4, -1\}$	Final answer.
		■

1.20

$$\frac{x+4}{x+2} - \frac{2x-3}{x-5} = \frac{3x-8}{x^2-3x-10}$$

	$\frac{x+4}{x+2} - \frac{2x-3}{x-5} = \frac{3x-8}{(x-5)(x+2)}$	Factor by grouping.
	$(x-5)(x+2)(\frac{x+4}{x+2} - \frac{2x-3}{x-5}) = (x-5)(x+2)(\frac{3x-8}{(x-5)(x+2)})$	lcm $= (x-5)(x+2)$
$x \notin \{5, -2\}$	$(x+4)(x-5) - (2x-3)(x+2) = 3x-8$	
$x \notin \{5, -2\}$	$x^2 - x - 20 - (2x^2 + x - 6) = 3x - 8$	
$x \notin \{5, -2\}$	$x^2 - x - 20 - 2x^2 - x + 6 = 3x - 8$	
$x \notin \{5, -2\}$	$x^2 - 2x^2 - x - x - 20 + 6 = 3x - 8$	
$x \notin \{5, -2\}$	$-x^2 - 2x - 14 = 3x - 8$	
$x \notin \{5, -2\}$	$-x^2 - 2x - 3x - 14 + 8 = 0$	Rewrite in standard form.
$x \notin \{5, -2\}$	$-x^2 - 5x - 6 = 0$	
$x \notin \{5, -2\}$	$x^2 + 5x + 6 = 0$	
$x \notin \{5, -2\}$	$(x+2)(x+3) = 0$	Factor by grouping.
$x \notin \{5, -2\}$	$x \in \{-2, -3\}$	
	$x = -3$	Final answer.
		■

1.21

$$1 + \frac{5x}{6} = \frac{2x^2}{3}$$

$6(1 + \frac{5x}{6}) = 6(\frac{2x^2}{3})$	lcm $= 6$
$6 + 5x = 4x^2$	
$4x^2 - 5x - 6 = 0$	Rewrite in standard form.
$4x^2 - 8x + 3x - 6 = 0$	Factor by grouping.
$(4x + 3)(x - 2) = 0$	
$x \in \{-\frac{3}{4}, 2\}$	Final answer.
	■

1.22

$$\frac{x}{2x+3} = x + 4$$

$x \neq -\frac{3}{2}$	$x = (2x+3)(x+4)$	Eliminate denominator.
$x \neq -\frac{3}{2}$	$x = 2x^2 + 11x + 12$	
$x \neq -\frac{3}{2}$	$2x^2 + 11x - x + 12 = 0$	Rewrite in standard form.
$x \neq -\frac{3}{2}$	$2x^2 + 10x + 12 = 0$	
$x \neq -\frac{3}{2}$	$x^2 + 5x + 6 = 0$	
$x \neq -\frac{3}{2}$	$(x+3)(x+2) = 0$	Factor by grouping.
	$x \in \{-3, -2\}$	Final answer.
		■

1.23

$$\sqrt{x} + 5 = 3$$

$\sqrt{x} = 3 - 5$	Isolate radical.
$\sqrt{x} = -2$	
$x = 4$	Eliminate radical.
$\sqrt{4} + 5 = 3$	Check $x = 4$.
$2 + 5 = 3$	
$7 \neq 3$	
$x \in \emptyset$	Final answer.
	■

1.24

$$\sqrt{3x+7} - x = 1$$

$\sqrt{3x+7} = x+1$	Isolate radical.
$3x+7 = (x+1)^2$	Eliminate radical.
$3x+7 = x^2+2x+1$	
$x^2+2x-3x+1-7 = 0$	Rewrite in standard form.
$x^2-x-6 = 0$	
$(x-3)(x+2) = 0$	Factor by grouping.
$x \in \{3, -2\}$	
$\sqrt{3(3)+7} = 3+1$	Check $x = 3$.
$\sqrt{9+7} = 3+1$	
$\sqrt{16} = 4$	
$4 = 4$	
$\sqrt{3(-2)+7} = 3+1$	Check $x = -2$.
$\sqrt{-6+7} = 3+1$	
$\sqrt{1} = 4$	
$1 \neq 4$	
$x = 3$	
	■

1.25

$$\sqrt{\sqrt{5x-1}-x}=1$$

$\sqrt{5x-1}-x=1$	Eliminate radical.
$\sqrt{5x-1}=x+1$	Isolate radical.
$5x-1=(x+1)^2$	Eliminate radical.
$5x-1=x^2+2x+1$	
$x^2+2x-5x+1+1=0$	Rewrite in standard form.
$x^2-3x+2=0$	
$(x-2)(x-1)=0$	Factor by grouping.
$x\in\{2,1\}$	
$\sqrt{\sqrt{5(2)-1}-2}=1$	Check $x=2$.
$\sqrt{\sqrt{9}-2}=1$	
$\sqrt{1}=1$	
$1=1$	
$\sqrt{\sqrt{5(1)-1}-1}=1$	Check $x-1$.
$\sqrt{\sqrt{4}-1}=1$	
$\sqrt{1}=1$	
$1=1$	
$x\in\{2,1\}$	Final answer.
	■

1.26

$$\sqrt{5w+1} - \sqrt{3w} - 1 = 0$$

$\sqrt{5w+1} - 1 = \sqrt{3w}$	Eliminate radicals.
$5w + 1 - 2\sqrt{5w+1} + 1 = 3w$	
$5w - 3w + 1 + 1 = 2\sqrt{5w+1}$	
$2w + 2 = 2\sqrt{5w+1}$	
$(2w+2)^2 = 4(5w+1)$	
$4w^2 + 8w + 4 = 20w + 4$	Rewrite in standard form.
$4w^2 + 8w - 20w + 4 - 4 = 0$	
$4w^2 - 12w = 0$	
$4w(w-3) = 0$	
$w \in \{0, 3\}$	
$\sqrt{5(0)+1} - \sqrt{3(0)} - 1 = 0$	Check $x = 0$.
$\sqrt{1} - \sqrt{0} - 1 = 0$	
$0 = 0$	
$\sqrt{5(3)+1} - \sqrt{3(3)} - 1 = 0$	Check $x = 3$.
$\sqrt{16} - \sqrt{9} - 1 = 0$	
$0 = 0$	
$w \in \{0, 3\}$	Final answer.
	■

1.27

$$\sqrt{x-2} = 2$$

$x - 2 = 4$ $x = 6$	Eliminate radical.
$\sqrt{6-2} = 2$ $2 = 2$	Check $x = 6$.
$x = 6$	Final answer. ■

1.28

$$\sqrt{2x+5}+x=5$$

$\sqrt{2x+5}=5-x$	Isolate radical.
$2x+5=(5-x)^2$	Eliminate radical.
$2x+5=25-10x+x^2$	
$x^2-10x-2x+25-5=0$	Rewrite in standard form.
$x^2-12x+20=0$	
$(x-10)(x-2)=0$	Factor by grouping.
$x\in\{10,2\}$	
$\sqrt{2(10)+5}+10=5$	Check $x=10$.
$\sqrt{20+5}+10=5$	
$15\neq 5$	
$\sqrt{2(2)+5}+2=5$	Check $x=2$.
$\sqrt{4+5}+2=5$	
$5=5$	
$x=2$	Final answer.
	■

1.29

$$\sqrt{x^2 - 8x} = 3$$

$x^2 - 8x = 9$ $x^2 - 8x - 9 = 0$ $(x+1)(x-9) = 0$ $x \in \{-1, 9\}$	Eliminate radical. Rewrite in standard form.
$\sqrt{(-1)^2 - 8(-1)} = 3$ $\sqrt{1+8} = 3$ $3 = 3$	Check $x = -1$.
$\sqrt{9^2 - 8(9)} = 3$ $\sqrt{81 - 72} = 3$ $3 = 3$	Check $x = 9$.
$x \in \{-1, 9\}$ ■	Final answer.

1.30

$$\sqrt{x+\sqrt{x-2}}=2$$

$x+\sqrt{x-2}=4$	Eliminate radical.
$\sqrt{x-2}=4-x$	Isolate radical.
$x-2=(4-x)^2$	Eliminate radical.
$x-2=16-8x+x^2$	
$x^2-8x-x+16+2=0$	Rewrite in standard form.
$x^2-9x+18=0$	
$(x-6)(x-3)=0$	
$x\in\{6,3\}$	
$\sqrt{6+\sqrt{6-2}}=2$	Check $x=6$.
$\sqrt{6+2}=2$	
$\sqrt{8}\neq 2$	
$\sqrt{3+\sqrt{3-2}}=2$	Check $x=3$.
$\sqrt{3+1}=2$	
	$2=2$
$x=3$	Final answer.
	■

1.31

$$\sqrt{x-1}-\sqrt{x-2}-1=0$$

$\sqrt{x-1}-1=\sqrt{x-2}$	Eliminate radicals.
$x-\cancel{1}-2\sqrt{x-1}+\cancel{1}=x-2$	
$\cancel{x}-\cancel{x}+2=2\sqrt{x-1}$	
$4=4(x-1)$	
$4=4x-4$	
$4x=8$	
$x=2$	
$\sqrt{2-1}-\sqrt{2-2}-1=0$	Check $x=2$.
$0=0$	
$x=2$	Final answer.
	■

1.32

$$2\sqrt{x-1}-\sqrt{x-2}-2=0$$

$2\sqrt{x-1}-2=\sqrt{x-2}$	Eliminate radicals.
$4(x-1)-8\sqrt{x-1}+4=x-2$	
$4x-\cancel{4}-8\sqrt{x-1}+\cancel{4}=x-2$	
$3x+2=8\sqrt{x-1}$	
$(3x+2)^2=64(x-1)$	
$9x^2+12x+4=64x-64$	
$9x^2+12x-64x+4+64=0$	
$9x^2-52x+68=0$	
$x=\frac{-(-52)\pm\sqrt{(-52)^2-4(9)(68)}}{2(9)}$	Use the quadratic formula.
$x=\frac{52\pm\sqrt{2704-2448}}{18}$	
$x=\frac{52\pm\sqrt{256}}{18}$	
$x=\frac{52\pm16}{18}$	
$x\in\{\frac{52+16}{18},\frac{52-16}{18}\}$	
$x\in\{\frac{68}{18},\frac{36}{18}\}$	
$x\in\{\frac{34}{9},2\}$	
$2\sqrt{\frac{34}{9}-1}-\sqrt{\frac{34}{9}-2}-2=0$	Check $x=\frac{34}{9}$.
$2\sqrt{\frac{25}{9}}-\sqrt{\frac{16}{9}}-2=0$	
$2(\frac{5}{3})-\frac{4}{3}-2=0$	
$\frac{10}{3}-\frac{4}{3}-2=0$	
$\frac{6}{3}-2=0$	
$0=0$	
$2\sqrt{2-1}-\sqrt{2-2}-2=0$	Check $x=2$.
$0=0$	
$x\in\{\frac{34}{9},2\}$	Final answer.
	■

1.33

$$\sqrt{3-3x}-\sqrt{3x+2}=3$$

$\sqrt{3-3x}-3=\sqrt{3x+2}$	Eliminate radicals.
$3-3x-6\sqrt{3-3x}+9=3x+2$	
$-3x-3x+3+9-2=6\sqrt{3-3x}$	
$-6x+10=6\sqrt{3-3x}$	
$(-6x+10)^2=36(3-3x)$	
$100-120x+36x^2=108-108x$	
$36x^2-120x+108x+100-108=0$	Rewrite in standard form.
$36x^2-12x-8=0$	
$9x^2-3x-2=0$	
$9x^2-6x+3x-2=0$	Factor by grouping.
$(3x-2)(3x+1)=0$	
$x=\{\frac{2}{3},-\frac{1}{3}\}$	
$\sqrt{3-3(\frac{2}{3})}-\sqrt{3(\frac{2}{3})+2}=3$	Check $x=\frac{2}{3}$.
$\sqrt{3-2}-\sqrt{2+2}=3$	
$-1\neq 3$	
$\sqrt{3-3(-\frac{1}{3})}-\sqrt{3(-\frac{1}{3})+2}=3$	Check $x=-\frac{1}{3}$.
$\sqrt{3+1}-\sqrt{-1+2}=3$	
$1\neq 3$	
$x\in\emptyset$	Final answer.
	■

1.34

$$2 + 2\sqrt[3]{x-1} = x$$

	$2\sqrt[3]{x-1} = x - 2$	Isolate radical.
	$8(x-1) = (x-2)^3$	
	$8x - \cancel{8} = x^3 - 6x^2 + 12x - \cancel{8}$	
	$x^3 - 6x^2 + 12x - 8x = 0$	Rewrite in standard form.
	$x^3 - 6x^2 + 4x = 0$	
	$x(x^2 - 6x + 4) = 0$	$x = 0$ may be a solution.
$x = 0$	$x = \frac{-(-6)\pm\sqrt{(-6)^2-4(1)(4)}}{2(1)}$	Find the zeroes of $x^2 - 6x + 4$ using the quadratic formula.
$x = 0$	$x = \frac{6\pm\sqrt{36-16}}{2}$	
$x = 0$	$x = \frac{6\pm\sqrt{20}}{2}$	
$x = 0$	$x = \frac{6\pm\sqrt{4}\sqrt{5}}{2}$	
$x = 0$	$x = \frac{6\pm2\sqrt{5}}{2}$	
	$x \in \{0, 3 \pm \sqrt{5}\}$	
	$2 + 2\sqrt[3]{0-1} = 0$	Check $x = 0$.
	$0 = 0$	
	$2 + 2\sqrt[3]{3+\sqrt{5}-1} = 3 + \sqrt{5}$	Check $x = 3 + \sqrt{5}$.
	$2 + 2\sqrt[3]{2+\sqrt{5}} = 3 + \sqrt{5}$	
	$2\sqrt[3]{2+\sqrt{5}} = 1 + \sqrt{5}$	
	$8(2+\sqrt{5}) = (1+\sqrt{5})^3$	
	$16 + 8\sqrt{5} = 1 + 3\sqrt{5} + 3(\sqrt{5})^2 + (\sqrt{5})^3$	$(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$
	$16 + 8\sqrt{5} = 1 + 3\sqrt{5} + 3(5) + 5\sqrt{5}$	
	$16 + 8\sqrt{5} = 1 + 15 + 3\sqrt{5} + 5\sqrt{5}$	
	$16 + 8\sqrt{5} = 16 + 8\sqrt{5}$	
	$2 + 2\sqrt[3]{3-\sqrt{5}-1} = 3 - \sqrt{5}$	Check $x = 3 - \sqrt{5}$.
	$2 + 2\sqrt[3]{2-\sqrt{5}} = 3 - \sqrt{5}$	
	$2\sqrt[3]{2-\sqrt{5}} = 1 - \sqrt{5}$	
	$8(2-\sqrt{5}) = (1-\sqrt{5})^3$	
	$16 - 8\sqrt{5} = 1 - 3\sqrt{5} + 3(\sqrt{5})^2 - (\sqrt{5})^3$	$(x-y)^3 = x^3 - 3x^2y + 3xy^2 - y^3$
	$16 - 8\sqrt{5} = 1 - 3\sqrt{5} + 3(5) - 5\sqrt{5}$	
	$16 - 8\sqrt{5} = 1 + 15 - 3\sqrt{5} - 5\sqrt{5}$	
	$16 - 8\sqrt{5} = 16 - 8\sqrt{5}$	

Continued on next page

$x \in \{0, 3 \pm \sqrt{5}\}$ Final answer.

■

1.35

$$\sqrt[4]{3x^2 - 3x + 1} = \sqrt{x - 2}$$

$3x^2 - 3x + 1 = (x-2)^2$	Eliminate radicals.
$3x^2 - 3x + 1 = x^2 - 4x + 4$	
$3x^2 - x^2 - 3x + 4x + 1 - 4 = 0$	Rewrite in standard form.
$2x^2 + x - 3 = 0$	
$2x^2 - 2x + 3x - 3 = 0$	Factor by grouping.
$(2x+3)(x-1) = 0$	
$x \in \{-\frac{3}{2}, 1\}$	
$\sqrt[4]{3(-\frac{3}{2})^2 - 3(-\frac{3}{2}) + 1} = \sqrt{-\frac{3}{2} - 2}$	Check $x = -\frac{3}{2}$.
$\sqrt[4]{3(\frac{9}{4}) - 3(-\frac{3}{2}) + 1} = \sqrt{-\frac{3}{2} - 2}$	
$\sqrt[4]{\frac{49}{4}} = \sqrt{-\frac{3}{2} - 2}$	
$\sqrt[4]{\frac{49}{4}} = \sqrt{-\frac{7}{2}}$	
$\sqrt{\frac{7}{2}} \neq i\sqrt{\frac{7}{2}}$	
$\sqrt[4]{3(1)^2 - 3(1) + 1} = \sqrt{1 - 2}$	Check $x = 1$.
$\sqrt[4]{1} = \sqrt{-1}$	
$1 \neq i$	
$x \in \emptyset$	Final answer.
	■

1.36

$$\sqrt{2x+3} - \sqrt{x-2} = \sqrt{x+1}$$

$2x + 3 - 2\sqrt{2x+3}\sqrt{x-2} + \cancel{x} - 2 = \cancel{x} + 1$	Eliminate radicals.
$2x - +3 - 2 - 1 = 2\sqrt{2x+3}\sqrt{x-2}$	
$2x = 2\sqrt{2x+3}\sqrt{x-2}$	
$4x^2 = 4(2x+3)(x-2)$	
$4x^2 = 4(2x^2 - x - 6)$	
$4x^2 = 8x^2 - 4x - 24$	
$8x^2 - 4x^2 - 4x - 24 = 0$	Rewrite in standard form.
$4x^2 - 4x - 24 = 0$	
$x^2 - x - 6 = 0$	
$(x-3)(x+2) = 0$	Factor by grouping.
$x \in \{3, -2\}$	
$\sqrt{2(3)+3} - \sqrt{3-2} = \sqrt{3+1}$	Check $x = 3$.
$\sqrt{9} - \sqrt{1} = \sqrt{4}$	
$2 = 2$	
$\sqrt{2(-2)+3} - \sqrt{-2-2} = \sqrt{-2+1}$	Check $x = -2$.
$\sqrt{-1} - \sqrt{-4} = \sqrt{-1}$	
$i - 2i = i$	
$-i \neq i$	
$x = 3$	Final answer.
	■

1.37

$$1 = x + \sqrt{2x - 3}$$

$1 - x = \sqrt{2x - 3}$	Isolate radical.
$(1 - x)^2 = 2x - 3$	Eliminate radical.
$1 - 2x + x^2 = 2x - 3$	
$x^2 - 2x - 2x + 1 + 3 = 0$	Rewrite in standard form.
$x^2 - 4x + 4 = 0$	
$(x - 2)^2 = 0$	
$x = 2$	
$1 = 2 + \sqrt{2(2) - 3}$	Check $x = 2$.
$1 = 2 + \sqrt{1}$	
$1 \neq 3$	
$x \in \emptyset$	Final answer.
	■

1.38

$$8(x+\frac{1}{2x})^2+30(x+\frac{1}{2x})+27=0$$

	$8t^2+30t+27=0$	$t=x+\frac{1}{2x}$
	$t=\frac{-30\pm\sqrt{30^2-4(8)(27)}}{2(8)}$	Use the quadratic formula.
	$t=\frac{-30\pm\sqrt{900-864}}{16}$	
	$t=\frac{-30\pm\sqrt{36}}{16}$	
	$t\in\{\frac{-30+6}{16},\frac{-30-6}{16}\}$	
	$t\in\{-\frac{24}{16},-\frac{36}{16}\}$	
	$t\in\{-\frac{3}{2},-\frac{9}{4}\}$	
	$-\frac{3}{2}=x+\frac{1}{2x}$	Solve for x, $t=-\frac{3}{2}$.
	$2x(-\frac{3}{2})=2x(x+\frac{1}{2x})$	lcm $=2x$
$x\neq 0$	$-3x=2x^2+1$	
$x\neq 0$	$2x^2+3x+1=0$	Rewrite in standard form.
$x\neq 0$	$2x^2+2x+x+1=0$	Factor by grouping.
$x\neq 0$	$(2x+1)(x+1)=0$	
	$x\in\{-\frac{1}{2},-1\}$	
	$-\frac{9}{4}=x+\frac{1}{2x}$	Solve for x, $t=-\frac{9}{4}$.
	$4x(-\frac{9}{4})=4x(x+\frac{1}{2x})$	lcm $=4x$
$x\neq 0$	$-9x=4x^2+2$	
$x\neq 0$	$4x^2+9x+2=0$	Rewrite in standard form.
$x\neq 0$	$4x^2+8x+x+2=0$	Factor by grouping.
$x\neq 0$	$(4x+1)(x+2)=0$	
	$x\in\{-\frac{1}{2},-1,-\frac{1}{4},-2\}$	
	$x\in\{-\frac{1}{2},-1,-\frac{1}{4},-2\}$	Final answer.
		■

1.39

$$\sqrt[4]{x} + 2\sqrt{x} = 3$$

$t + 2t^2 = 3$	$t = \sqrt[4]{x}$
$2t^2 + t - 3 = 0$	Rewrite in standard form.
$2t^2 + 3t - 2t - 3 = 0$	Factor by grouping.
$(t-1)(2t+3) = 0$	
$t \in \{1, -\frac{3}{2}\}$	
$1 = \sqrt[4]{x}$	Solve for x, $t = 1$.
$x = 1$	
$1 = \sqrt[4]{1}$	Check $x = 1$.
$1 = 1$	
$-\frac{3}{2} = \sqrt[4]{x}$	Solve for x, $t = -\frac{3}{2}$.
$x = \frac{81}{16}$	
$-\frac{3}{2} = \sqrt[4]{\frac{81}{16}}$	Check $x = \frac{81}{16}$.
$-\frac{3}{2} \neq \frac{3}{2}$	
$x = 1$	Final answer.
	■

1.40

$$\sqrt{x} - 5\sqrt[4]{x} + 4 = 0$$

$t^2 - 5t + 4 = 0$ $(t-4)(t-1) = 0$ $t \in \{4, 1\}$	$t = \sqrt[4]{x}$ Factor by grouping.
$4 = \sqrt[4]{x}$ $x = 256$	Solve for x, $t = 4$.
$4 = \sqrt[4]{256}$ $4 = 4$	Check $x = 256$.
$1 = \sqrt[4]{x}$ $x = 1$	Solve for x, $t = 1$.
$4 = \sqrt[4]{1}$ $1 = 1$	Check $x = 1$.
$x \in \{256, 1\}$ ■	Final answer.

1.41

$$\frac{15}{t^2} - \frac{14}{t} - 8 = 0$$

	$t^2(\frac{15}{t^2} - \frac{14}{t} - 8) = t^2(0)$	lcm $= t^2$
$t \neq 0$	$15 - 14t - 8t^2 = 0$	
$t \neq 0$	$8t^2 + 14t - 15 = 0$	Rewrite in standard form.
$t \neq 0$	$t = \frac{-14 \pm \sqrt{14^2 - 4(8)(-15)}}{2(8)}$	Use the quadratic formula.
$t \neq 0$	$t = \frac{-14 \pm \sqrt{196+480}}{16}$	
$t \neq 0$	$t = \frac{-14 \pm \sqrt{676}}{16}$	
$t \neq 0$	$t = \frac{-14 \pm 26}{16}$	
$t \neq 0$	$t \in \{\frac{-14+26}{16}, \frac{-14+26}{16}\}$	
$t \neq 0$	$t \in \{\frac{12}{16}, -\frac{40}{16}\}$	
	$t \in \{\frac{3}{4}, -\frac{5}{2}\}$	Final answer.
		■

1.42

$$t^4 - 4 = 0$$

	$(t^2-2)(t^2+2)=0$	Factor using difference of two squares.
	$(t-\sqrt{2})(t+\sqrt{2})(t^2+2)=0$	Factor using difference of two squares. $\pm\sqrt{2}$ are solutions.
$t=\pm\sqrt{2}$	$t^2+2=0$	Find the zeroes of t^2+2.
$t=\pm\sqrt{2}$	$t^2=-2$	
$t=\pm\sqrt{2}$	$t=\pm\sqrt{-2}$	
$t=\pm\sqrt{2}$	$t=\pm i\sqrt{2}$	
	$t \in \{\pm\sqrt{2}, \pm i\sqrt{2}\}$	Final answer.
		■

1.43

$$x^4 - 2x^2 - 15 = 0$$

$t^2 - 2t - 15 = 0$ $(t-5)(t+3) = 0$ $t \in \{5, 3\}$	$t = x^2$ Factor by grouping.
$5 = x^2$ $x = \pm\sqrt{5}$	Solve for x, $t = 5$.
$-3 = x^2$ $x = \pm i\sqrt{3}$	Solve for x, $t = -3$.
$x \in \{\pm\sqrt{5}, \pm i\sqrt{3}\}$	Final answer. ■

1.44

$$2(2x+\frac{3}{2})^2 - 7(2x+\frac{3}{2}) + 3 = 0$$

$2t^2 - 7t + 3 = 0$ $2t^2 - 6t - t + 3 = 0$ $(2t-1)(t-3) = 0$ $t \in \{\frac{1}{2}, 3\}$	$t = 2x + \frac{3}{2}$ Factor by grouping.
$\frac{1}{2} = 2x + \frac{3}{2}$ $\frac{1}{2} - \frac{3}{2} = 2x$ $-1 = 2x$ $x = -\frac{1}{2}$	Solve for x, $t = \frac{1}{2}$.
$3 = 2x + \frac{3}{2}$ $3 - \frac{3}{2} = 2x$ $\frac{3}{2} = 2x$ $x = \frac{3}{4}$	Solve for x, $t = 3$.
$x \in \{-\frac{1}{2}, \frac{3}{4}\}$ ■	Final answer.

1.45

$$2x^{\frac{2}{3}} - 5x^{\frac{1}{3}} - 3 = 0$$

$2t^2 - 5t - 3 = 0$	$t = x^{\frac{1}{3}}$
$2t^2 - 6t + t - 3 = 0$	Factor by grouping.
$(2t+1)(t-3) = 0$	
$t \in \{-\frac{1}{2}, 3\}$	
$-\frac{1}{2} = x^{\frac{1}{3}}$	Solve for x, $t = -\frac{1}{2}$.
$(-\frac{1}{2})^3 = x$	
$x = -\frac{1}{8}$	
$3 = x^{\frac{1}{3}}$	Solve for x, $t = 3$.
$3^3 = x$	
$x = 27$	
$x \in \{-\frac{1}{8}, 27\}$	Final answer.
	■

1.46

$$3(4x - \frac{1}{4})^2 - 4(4x - \frac{1}{4}) - 15 = 0$$

$3t^2 - 4t - 15 = 0$	$t = 4x - \frac{1}{4}$
$3t^2 - 9t + 5t - 15 = 0$	Factor by grouping.
$(3t + 5)(t - 3) = 0$	
$t \in \{-\frac{5}{3}, 3\}$	
$-\frac{5}{3} = 4x - \frac{1}{4}$	Solve for x, $t = -\frac{5}{3}$.
$-\frac{5}{3} + \frac{1}{4} = 4x$	
$-\frac{20}{12} + \frac{3}{12} = 4x$	
$-\frac{17}{12} = 4x$	
$x = -\frac{17}{48}$	
$3 = 4x - \frac{1}{4}$	Solve for x, $t = 3$.
$3 + \frac{1}{4} = 4x$	
$\frac{13}{4} = 4x$	
$x = \frac{13}{16}$	
$x \in \{-\frac{17}{48}, \frac{13}{16}\}$	Final answer.
	■

1.47

$$6 + \sqrt[3]{x+1} = \sqrt[3]{(x+1)^2}$$

$6 + t = t^2$	$t = \sqrt[3]{x+1}$
$t^2 - t - 6 = 0$	Rewrite in standard form.
$(t-3)(t+2) = 0$	Factor by grouping.
$t \in \{3, -2\}$	
$3 = \sqrt[3]{x+1}$	Solve for x, $t = 3$.
$3^3 = x + 1$	Eliminate radical.
$27 = x + 1$	
$x = 26$	
$3 = \sqrt[3]{26+1}$	Check $x = 26$.
$3 = 3$	
$-2 = \sqrt[3]{x+1}$	Solve for x, $t = -2$.
$(-2)^3 = x + 1$	Eliminate radical.
$-8 = x + 1$	
$x = -9$	
$-2 = \sqrt[3]{-9+1}$	Check $x = -9$.
$-2 = -2$	
$x \in \{26, -9\}$	Final answer.
	■

1.48

$$-7(\frac{1}{x}-1) = 4 - 2(\frac{1}{x}-1)^2$$

	$-7t = 4 - 2t^2$	$t = \frac{1}{x} - 1$
	$2t^2 - 7t - 4 = 0$	Rewrite in standard form.
	$2t^2 - 8t + t - 4 = 0$	Factor by grouping.
	$(2t+1)(t-4) = 0$	
	$t \in \{-\frac{1}{2}, 4\}$	
	$-\frac{1}{2} = \frac{1}{x} - 1$	Solve for x, $t = -\frac{1}{2}$.
	$2x(-\frac{1}{2}) = 2x(\frac{1}{x} - 1)$	lcm $= 2x$
$x \neq 0$	$-x = 2 - 2x$	
	$x = 2$	
	$4 = \frac{1}{x} - 1$	Solve for x, $t = 4$.
$x \neq 0$	$4x = 1 - x$	lcm $= x$
$x \neq 0$	$5x = 1$	
	$x = \frac{1}{5}$	
	$x \in \{2, \frac{1}{5}\}$	Final answer.
		■

1.49

$$x^2(x^2-1)-9(x^2-1)=0$$

$x^4-x^2-9x^2+9=0$	
$x^4-10x^2+9=0$	
$t^2-10t+9=0$	$t=x^2$
$(t-9)(t-1)=0$	Factor by grouping.
$t\in\{9,1\}$	
$9=x^2$	Solve for x, $t=9$.
$x=\pm 3$	
$1=x^2$	Solve for x, $t=1$.
$x=\pm 1$	
$x\in\{\pm 3,\pm 1\}$	Final answer.
	■

1.50

$$2(x^2+x+1)+\sqrt{x^2+x+1}-3=0$$

$2t^2+t-3=0$	$t=\sqrt{x^2+x+1}$
$2t^2+3t-2t-3=0$	Factor by grouping.
$(t-1)(2t+3)=0$	
$t\in\{1,-\frac{3}{2}\}$	
$1=\sqrt{x^2+x+1}$	Solve for x, $t=1$.
$1=x^2+x+1$	Eliminate radical.
$x^2+x+1-1=0$	Rewrite in standard form.
$x^2+x=0$	
$x(x+1)=0$	
$x\in\{0,-1\}$	
$1=\sqrt{0^2+0+1}$	Check $x=0$.
$1=1$	
$1=\sqrt{(-1)^2+(-1)+1}$	Check $x=-1$.
$1=1$	
$-\frac{3}{2}=\sqrt{x^2+x+1}$	Solve for x, $t=-\frac{3}{2}$.
$x\in\emptyset$	The square root of a real number cannot be negative.
$x\in\{0,1\}$	Final answer.
	■

1.51

$$|7 - 2w| = 9$$

$7 - 2w = 9$ $-2w = 9 - 7$ $-2w = 2$ $w = -1$	Solve positive case.
$7 - 2w = -9$ $-2w = -9 - 7$ $-2w = -16$ $w = 8$ $w \in \{-1, 8\}$ ■	Solve negative case.

1.52

$$|3x-2| = |2x+3|$$

$3x-2=2x+3$ $3x-2x=3+2$ $x=5$	Solve positive case.
$3x-2=-(2x+3)$ $3x-2=-2x-3$ $3x+2x=-3+2$ $5x=-1$ $x=-\frac{1}{5}$	Solve negative case.
$\lvert 3(5)-2\rvert = \lvert 2(5)+3\rvert$ $\lvert 15-2\rvert = \lvert 10+3\rvert$ $13=13$	Check $x=5$.
$\left\lvert 3(-\frac{1}{5})-2\right\rvert = \left\lvert 2(-\frac{1}{5})+3\right\rvert$ $\left\lvert -\frac{3}{5}-2\right\rvert = \left\lvert -\frac{2}{5}+3\right\rvert$ $\left\lvert -\frac{3}{5}-\frac{10}{5}\right\rvert = \left\lvert -\frac{2}{5}+\frac{15}{5}\right\rvert$ $\frac{13}{5}=\frac{13}{5}$	Check $x=-\frac{1}{5}$.
$x \in \{5, -\frac{1}{5}\}$ ■	Final answer.

1.53

$$|x-4| = |5-2x|$$

$x-4=5-2x$ $x+2x=5+4$ $3x=9$ $x=3$	Solve positive case.
$x-4=-(5-2x)$ $x-4=2x-5$ $x-2x=-5+4$ $x=1$	Solve negative case.
$\|3-4\| = \|5-2(3)\|$ $\|3-4\| = \|5-6\|$ $1=1$	Check $x=3$.
$\|1-4\| = \|5-2(1)\|$ $\|1-4\| = \|5-2\|$ $3=3$	Check $x=1$.
$x \in \{1,3\}$ ■	Final answer.

1.54

$$\left|\frac{x+3}{x-3}\right| = 7$$

	$\frac{x+3}{x-3} = 7$	Solve positive case.
$x \neq 3$	$x + 3 = 7(x - 3)$	
$x \neq 3$	$x + 3 = 7x - 21$	
$x \neq 3$	$x - 7x = -21 - 3$	
$x \neq 3$	$-6x = -24$	
	$x = 4$	
	$\frac{x+3}{x-3} = 7$	Solve negative case.
$x \neq 3$	$x + 3 = -7(x - 3)$	
$x \neq 3$	$x + 3 = -7x + 21$	
$x \neq 3$	$x + 7x = 21 - 3$	
$x \neq 3$	$8x = 18$	
$x \neq 3$	$x = \frac{18}{8}$	
	$x = \frac{9}{4}$	
	$x \in \{4, \frac{9}{4}\}$	Final answer.
		■

1.55

$$\left|\frac{2x+1}{x-1}\right| = 3$$

	$\frac{2x+1}{x-1} = 3$	Solve positive case.
$x \neq 1$	$2x+1 = 3(x-1)$	
$x \neq 1$	$2x+1 = 3x-3$	
$x \neq 1$	$2x-3x = -3-1$	
	$x = 4$	
	$\frac{2x+1}{x-1} = -3$	Solve negativee case.
$x \neq 1$	$2x+1 = -3(x-1)$	
$x \neq 1$	$2x+1 = -3x+3$	
$x \neq 1$	$2x+3x = 3-1$	
$x \neq 1$	$5x = 2$	
	$x = \frac{2}{5}$	
	$x \in \{4, \frac{2}{5}\}$	Final answer.
		■

1.56

$$|3t-2| = t^2$$

$3t-2=t^2$	Solve positive case.
$t^2-3t+2=0$	Rewrite in standard form.
$(t-2)(t-1)=0$	Factor by grouping.
$t \in \{2,1\}$	
$3t-2=-t^2$	Solve negative case.
$t^2+3t-2=0$	Rewrite in standard form.
$t=\frac{-3\pm\sqrt{3^2-4(1)(-2)}}{2(1)}$	Use the quadratic formula.
$t=\frac{-3\pm\sqrt{9+8}}{2}$	
$t=\frac{-3\pm\sqrt{17}}{2}$	
$t \in \{2,1,\frac{-3\pm\sqrt{17}}{2}\}$	Final answer.
	■

1.57

$$|3t - 2| = -t^2$$

$t \in \emptyset$	Final answer. Since t^2 will always be positive, $-t^2$ will always be negative. The absolute value of a real number cannot be negative. ■

1.58

$$|3x+5| = 9$$

$3x+5=9$ $3x=9-5$ $3x=4$ $x=\frac{4}{3}$	Solve positive case.
$3x+5=-9$ $3x=-9-5$ $3x=-14$ $x=-\frac{14}{3}$	Solve negative case.
$x \in \{\frac{4}{3}, -\frac{14}{3}\}$ ■	Final answer.

1.59

$$|-3x+7|-3=0$$

$\lvert -3x+7\rvert = 3$	
$-3x+7=3$ $-3x=3-7$ $-3x=-4$ $x=\frac{4}{3}$	Solve positive case.
$-3x+7=-3$ $-3x=-3-7$ $-3x=-10$ $x=\frac{10}{3}$	Solve negative case.
$x \in \{\frac{4}{3}, \frac{10}{3}\}$ ■	Final answer.

1.60

$$\left|\frac{x}{2x-3}\right| = 2$$

	$\frac{x}{2x-3} = 2$	Solve positive case.
$x \neq \frac{3}{2}$	$x = 2(2x-3)$	
$x \neq \frac{3}{2}$	$x = 4x - 6$	
$x \neq \frac{3}{2}$	$x - 4x = -6$	
$x \neq \frac{3}{2}$	$-3x = -6$	
	$x = 2$	
	$\frac{x}{2x-3} = -2$	Solve negative case.
$x \neq \frac{3}{2}$	$x = -2(2x-3)$	
$x \neq \frac{3}{2}$	$x = -4x + 6$	
$x \neq \frac{3}{2}$	$x + 4x = 6$	
$x \neq \frac{3}{2}$	$5x = 6$	
	$x = \frac{6}{5}$	
	$x \in \{2, \frac{6}{5}\}$	Final answer.
		■

1.61

$$\left|\frac{3x-4}{2x+3}\right| = 1$$

	$\frac{3x-4}{2x+3} = 1$	Solve positive case.
$x \neq -\frac{3}{2}$	$3x - 4 = 2x + 3$	
$x \neq -\frac{3}{2}$	$3x - 2x = 3 + 4$	
	$x = 7$	
	$\frac{3x-4}{2x+3} = -1$	Solve negative case.
$x \neq -\frac{3}{2}$	$3x - 4 = -1(2x + 3)$	
$x \neq -\frac{3}{2}$	$3x - 4 = -2x - 3$	
$x \neq -\frac{3}{2}$	$3x + 2x = -3 + 4$	
$x \neq -\frac{3}{2}$	$5x = 1$	
	$x = \frac{1}{5}$	
	$x \in \{7, \frac{1}{5}\}$	Final answer.
		■

1.62

$$|2x-3| = |7-3x|$$

$2x-3=7-3x$ $2x+3x=7+3$ $5x=10$ $x=2$	Solve positive case.
$2x-3=-(7-3x)$ $2x-3=-7+3x$ $2x-3x=-7+3$ $x=4$	Solve negative case.
$\lvert 2(2)-3\rvert = \lvert 7-3(2)\rvert$ $\lvert 4-3\rvert = \lvert 7-6\rvert$ $1=1$	Check $x=2$.
$\lvert 2(4)-3\rvert = \lvert 7-3(4)\rvert$ $\lvert 8-3\rvert = \lvert 7-12\rvert$ $5=5$	Check $x=4$.
$x \in \{2,4\}$ ■	Final answer.

1.63

$$|4x - 3| = x$$

$4x - 3 = x$ $3x = 3$ $x = 1$	Solve positive case.
$4x - 3 = -x$ $5x = 3$ $x = \frac{3}{5}$	Solve negative case.
$x \in \{1, \frac{3}{5}\}$ ■	Final answer.

1.64

$$|3x + 8| = x$$

$3x + 8 = x$ $2x = -8$ $x = -4$	Solve positive case.
$3x + 8 = -x$ $4x = -8$ $x = -2$	Solve negative case.
$x \in \emptyset$	Final answer. x cannot be negative; the absolute value of a real number cannot be negative. ■

1.65

$$|4x - 3| = -x$$

$4x - 3 = -x$ $5x = 3$ $x = \frac{3}{5}$	Solve positive case.
$4x - 3 = -(-x)$ $3x = 3$ $x = 1$	Solve negative case.
$x \in \emptyset$	Final answer. x cannot be positive because $-x$ would be negative; the absolute value of a real number cannot be negative. ■

1.66

$$|3x + 8| = -x$$

$3x + 8 = -x$ $4x = -8$ $x = -2$	Solve negative case.
$3x + 8 = -(-x)$ $2x = -8$ $x = -4$	Solve positive case.
$x \in \{-2, -4\}$ ■	Final answer.

2

Find all real values of k such that there is exactly one solution to the equation.

2.1

$$x^2 - kx + k = x - 2$$

$x^2 - kx - x + k + 2 = 0$ $x^2 - (k+1)x + (k+2) = 0$	Rewrite in standard form.
$(k+1)^2 - 4(1)(k+2) = 0$ $k^2 + 2k + 1 - 4k - 8 = 0$ $k^2 - 2k - 7 = 0$ $k = \frac{-(-2)\pm\sqrt{(-2)^2-4(1)(-7)}}{2(1)}$ $k = \frac{2\pm\sqrt{4+28}}{2}$ $k = \frac{2\pm\sqrt{32}}{2}$ $k = \frac{2\pm4\sqrt{2}}{2}$	If the determinant $b^2 - 4ac$ is 0, the quadratic equation only has one solution. Use the quadratic formula.
$k = 1 \pm 2\sqrt{2}$ ■	Final answer.

3

Find the solution set of the following inequalities.

3.1

$$x - 2 < 4 - x$$

$2x < 6$ $x < 3$	
$x \in (-\infty, 3)$	Final answer.
	■

3.2

$$x + 1 > \frac{2x - 4}{5}$$

$5x + 5 > 2x - 4$ $5x - 2x > -4 - 5$ $3x > -9$ $x > -3$	
$x \in (-3, +\infty)$	Final answer.
	■

3.3

$$\frac{2x-3}{4} \geq \frac{7x}{2}$$

$4(\frac{2x-3}{4}) \geq 4(\frac{7x}{2})$	lcm = 4
$2x - 3 \geq 14x$	
$12x \leq -3$	
$x \leq -\frac{3}{12}$	
$x \leq -\frac{1}{4}$	
$x \in (-\infty, -\frac{1}{4}]$	Final answer.
	■

3.4

$$-2 < 5 + 3x < 20$$

$-7 < 3x < 15$ $-\frac{7}{3} < x < 5$	
$x \in (-\frac{7}{3}, 5)$	Final answer. ■

3.5

$$5 \leq 3x - 4 \leq 7$$

$9 \leq 3x \leq 11$ $3 \leq x \leq \frac{11}{3}$	
$x \in [3, \frac{11}{3}]$	Final answer.
	■

3.6

$$2x - 1 \leq 3x \leq 4x + 1$$

$2x - 1 \leq 3x$ $x \geq -1$	Solve left inequality.
$3x \leq 4x + 1$ $x \geq -1$	Solve right inequality.
$x \geq -1$	Find the intersection of the inequalities.
$x \in [-1, +\infty)$ ■	Final answer.

3.7

$$3x^2 \leq 11x - 10$$

$3x^2 - 11x + 10 \leq 0$

$3x^2 - 5x - 6x + 10 \leq 0$ Factor by grouping.

$(x-2)(3x-5) \leq 0$

$N_C \in \{\frac{5}{3}, 2\}$

Create a table of signs.

		$\frac{5}{3}$		2	
$3x - 5$	$-$		$+$		$+$
$x - 2$	$-$		$-$		$+$
$(x-2)(3x-5)$	$+$		$-$		$+$

$x \in [\frac{5}{3}, 2]$ Final answer.

■

3.8

$$(x+3)(x-4) \geq 3x$$

$x^2 - x - 12 \geq 3x$

$x^2 - 4x - 12 \geq 0$

$(x+2)(x-6) \geq 0$ — Factor by grouping.

$N_C \in \{-2, 6\}$

Create a table of signs.

		-2		6	
$x + 2$	$-$		$+$		$+$
$x - 6$	$-$		$-$		$+$
$(x+2)(x-6)$	$+$		$-$		$+$

$x \in (-\infty, -2] \cup [6, +\infty)$ — Final answer.

■

3.9

$$x^3 + x \geq x^2 + 1$$

$x^3 - x^2 + x - 1 \geq 0$	
$(x^2 + 1)(x - 1) \geq 0$	Factor by grouping.
$x \geq 1$	$x^2 + 1$ is always positive. Evaluate $x - 1 \geq 0$.
$x \in [1, +\infty)$	Final answer.
	■

3.10

$$x^2 > 9$$

$x^2 - 9 > 0$

$(x-3)(x+3) > 0$ — Factor using difference of two squares.

$N_C \in \{-3, 3\}$

Create a table of signs.

	-3	3	
$x + 3$	$-$	$+$	$+$
$x - 3$	$-$	$-$	$+$
$(x+3)(x-3)$	$+$	$-$	$+$

$x \in (-\infty, -3) \cup (3, +\infty)$ — Final answer.

■

3.11

$$4 - 3x - x^2 \geq 0$$

$x^2 + 3x - 4 \leq 0$

$(x+4)(x-1) \leq 0$ — Factor by grouping.

$N_C \in \{-4, 1\}$

Create a table of signs.

	-4		1
$x + 4$	$-$	$+$	$+$
$x - 1$	$-$	$-$	$+$
$(x+4)(x-1)$	$+$	$-$	$+$

$x \in [-4, 1]$ — Final answer.

■

3.12

$$4x^2 + 4x + 1 < 0$$

$4x^2 + 2x + 2x + 1 < 0$ $(2x+1)^2 < 0$	Factor by grouping.
$x \in \emptyset$	Final answer. The square of a real number cannot be less than zero. ■

3.13

$$4x^2 + 4x > -1$$

$4x^2 + 4x + 1 > 0$	
$(2x+1)^2 > 0$	Factor by grouping.
$x \in \mathbb{R} \setminus \{-\frac{1}{2}\}$	Final answer. The square of a real number is always greater than or equal to zero. $x = -\frac{1}{2}$ is not included since then the inequality will become $0 > 0$. ■

3.14

$$x^2 - 2x < 2x - 3$$

$x^2 - 4x + 3 < 0$

$(x-1)(x-3) < 0$ — Factor by grouping.

$N_C \in \{1, 3\}$

Create a table of signs.

		1		3	
$x - 1$	$-$		$+$		$+$
$x - 3$	$-$		$-$		$+$
$(x-1)(x-3)$	$+$		$-$		$+$

$x \in (1, 3)$ — Final answer.

■

3.15

$$(x+4)(x^2-4) < 0$$

$(x+4)(x+2)(x-2) < 0$ — Factor using difference of two squares.

$N_C \in \{-4, -2, 2\}$

Create a table of signs.

	-4	-2	2	
$x + 4$	$-$	$+$	$+$	$+$
$x + 2$	$-$	$-$	$+$	$+$
$x - 2$	$-$	$-$	$-$	$+$
$(x+4)(x+2)(x-2)$	$-$	$+$	$-$	$+$

$x \in (-\infty, -4) \cup (-2, 2)$ — Final answer.

■

3.16

$$x - 2 \geq (x-2)(2x^2 - x - 14)$$

$x - 2 \geq 2x^3 - x^2 - 14x - 4x^2 + 2x + 28$

$x - 2 \geq 2x^3 - 5x^2 - 12x + 28$

$2x^3 - 5x^2 - 13x + 30 \leq 0$

$p \in \{\pm 1, \pm 2, \pm 3, \pm 5, \pm 10, \pm 15, \pm 30\}$ — Factor using rational root theorem.

$q \in \{\pm 1, \pm 2\}$

$\frac{p}{q} \in \{\pm 1, \pm \frac{1}{2}, \pm 2, \pm 3, \pm \frac{3}{2}, \pm 5, \pm \frac{5}{2}, \pm 10, \pm 15, \pm \frac{15}{2}, \pm 30\}$

Try $x = 1$.

$$\begin{array}{r|rrrr} & 2 & -5 & -13 & 30 \\ 1 & & 2 & -3 & -16 \\ \hline & 2 & -3 & -16 & 14 \end{array}$$

Try x = -1.

$$\begin{array}{r|rrrr} & 2 & -5 & -13 & 30 \\ -1 & & -2 & 7 & 6 \\ \hline & 2 & -7 & -6 & 36 \end{array}$$

Try $x = 2$.

$$\begin{array}{r|rrrr} & 2 & -5 & -13 & 30 \\ 2 & & 4 & -2 & -30 \\ \hline & 2 & -1 & -15 & 0 \end{array}$$

$(x-2)(2x^2 - x - 15) \leq 0$

$(x-2)(2x^2 - 6x + 5x - 15) \leq 0$ — Factor by grouping.

$(x-2)(2x+5)(x-3) \leq 0$

$N_C \in \{-\frac{5}{2}, 2, 3\}$

Create a table of signs.

		$-\frac{5}{2}$		2		3	
$2x + 5$	$-$		$+$		$+$		$+$
$x - 2$	$-$		$-$		$+$		$+$
$x - 3$	$-$		$-$		$-$		$+$
$(x-2)(2x+5)(x-3)$	$-$		$+$		$-$		$+$

$x \in (-\infty, -\frac{5}{2}] \cup [2, 3]$ — Final answer.

■

3.17

$$x(2x^2 - 5x - 12) \geq 2x^2 - 5x - 12$$

$2x^3 - 5x^2 - 12x \geq 2x^2 - 5x - 12$

$2x^3 - 7x^2 - 7x + 12 \geq 0$

$p \in \{\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12\}$ — Factor using rational root theorem.

$q \in \{\pm 1, \pm 2\}$

$\frac{p}{q} \in \{\pm 1, \pm \frac{1}{2}, \pm 2, \pm 3, \pm \frac{3}{2}, \pm 4, \pm 6, \pm 12\}$

Try $x = 1$.

$$\begin{array}{r|rrrr} & 2 & -7 & -7 & 12 \\ 1 & & 2 & -5 & -12 \\ \hline & 2 & -5 & -12 & 0 \end{array}$$

$(x-1)(2x^2 - 5x - 12) \geq 0$

$(x-1)(2x^2 - 8x + 3x - 12) \geq 0$ — Factor by grouping.

$(x-1)(2x+3)(x-4) \geq 0$

$N_C \in \{-\frac{3}{2}, 1, 4\}$

Create a table of signs.

	$-\frac{3}{2}$	1	4	
$2x + 3$	$-$	$+$	$+$	$+$
$x - 1$	$-$	$-$	$+$	$+$
$x - 4$	$-$	$-$	$-$	$+$
$(x-1)(2x+3)(x-4)$	$-$	$+$	$-$	$+$

$x \in [-\frac{3}{2}, 1] \cup [4, +\infty)$ — Final answer.

■

3.18

$$\frac{x}{x-1} > -1$$

$\frac{x}{x-1} + \frac{x-1}{x-1} > 0$

$\frac{2x-1}{x-1} > 0$

$x \neq 1 \quad N_C \in \{\frac{1}{2}, 1\}$

Create a table of signs.

		$\frac{1}{2}$		1	
$2x-1$	$-$		$+$		$+$
$x-1$	$-$		$-$		$+$
$\frac{2x-1}{x-1}$	$+$		$-$	$\bullet$	$+$

$x \in (-\infty, \frac{1}{2}) \cup (1, +\infty)$ Final answer.

■

3.19

$$\frac{x-1}{2x+5} \geq 0$$

$x \neq -\frac{5}{2}$ $\quad N_C \in \{-\frac{5}{2}, 1\}$

Create a table of signs.

		$-\frac{5}{2}$		1	
$2x+5$	$-$		$+$		$+$
$x-1$	$-$		$-$		$+$
$\frac{x-1}{2x+5}$	$+$	$\bullet$	$-$		$+$

$x \in (-\infty, -\frac{5}{2}) \cup [1, +\infty)$ Final answer.

■

3.20

$$\frac{6-2x}{4+x} > 5$$

$\frac{6-2x}{4+x} - \frac{20+5x}{4+x} > 0$

$\frac{-7x-14}{4+x} > 0$

$\frac{-x-2}{4+x} > 0$

$\frac{x+2}{4+x} < 0$

$x \neq -4 \quad N_C \in \{-4, -2\}$

Create a table of signs.

		-4		-2	
$4+x$	$-$		$+$		$+$
$x+2$	$-$		$-$		$+$
$\frac{x+2}{4+x}$	$+$	$\bullet$	$-$		$+$

$x \in (-4, -2)$

Final answer.

■

3.21

$$\frac{x^2 - x - 2}{x} \leq 0$$

$\frac{(x-2)(x+1)}{x} \leq 0$ — Factor by grouping.

$x \neq 0 \quad N_C \in \{-1, 0, 2\}$

Create a table of signs.

		-1		0		2	
$x+1$	$-$		$+$		$+$		$+$
x	$-$		$-$		$+$		$+$
$x-2$	$-$		$-$		$-$		$+$
$\frac{(x-2)(x+1)}{x}$	$-$		$+$	$\bullet$	$-$		$+$

$x \in (-\infty, -1] \cup (0, 2]$ — Final answer.

■

3.22

$$\frac{3x-1}{x^2-x-6} \leq 1$$

$\frac{3x-1}{x^2-x-6} - \frac{x^2-x-6}{x^2-x-6} \leq 0$

$\frac{-x^2+4x+5}{x^2-x-6} \leq 0$

$\frac{x^2-4x-5}{x^2-x-6} \geq 0$

$\frac{(x-5)(x+1)}{(x-3)(x+2)} \geq 0$ Factor by grouping.

$x \notin \{-2, 3\}$ $N_C \in \{-2, -1, 3, 5\}$

Create a table of signs.

		-2		-1		3		5	
$x+2$	$-$		$+$		$+$		$+$		$+$
$x+1$	$-$		$-$		$+$		$+$		$+$
$x-3$	$-$		$-$		$-$		$+$		$+$
$x-5$	$-$		$-$		$-$		$-$		$+$
$\frac{(x-5)(x+1)}{(x-3)(x+2)}$	$+$	$\bullet$	$-$		$+$	$\bullet$	$-$		$+$

$x \in (-\infty, -2) \cup [-1, 3) \cup [5, +\infty)$ Final answer.

■

3.23

$$\frac{x}{x-2} \leq 1$$

	$\frac{x}{x-2} - \frac{x-2}{x-2} \leq 0$	
	$\frac{2}{x-2} \leq 0$	
$x \neq 2$	$x \leq 2$	
	$x \in (-\infty, 2)$	Final answer.
		■

3.24

$$\frac{1}{x-2} \leq \frac{1}{x}$$

$\frac{1}{x-2} - \frac{1}{x} \leq 0$

$\frac{x}{x(x-2)} - \frac{x-2}{x(x-2)} \leq 0$ lcm $= x(x-2)$

$\frac{2}{x(x-2)} \leq 0$

$x \notin \{0, 2\}$ $N_C \in \{0, 2\}$

Create a table of signs.

		0		2	
x	$-$		$+$		$+$
$x-2$	$-$		$-$		$+$
$\frac{2}{x(x-2)}$	$+$	$\bullet$	$-$	$\bullet$	$+$

$x \in (0, 2)$ Final answer.

■

3.25

$$\frac{1}{x-2} \leq \frac{1}{3}$$

$\frac{1}{x-2} - \frac{1}{3} \leq 0$

$\frac{3}{3(x-2)} - \frac{x-2}{3(x-2)} \leq 0$ — lcm $= 3(x-2)$

$\frac{-x+5}{3(x-2)} \leq 0$

$\frac{x-5}{3(x-2)} \geq 0$

$x \neq 2 \quad N_C \in \{2, 5\}$

Create a table of signs.

		2		5	
$x-2$	$-$		$+$		$+$
$x-5$	$-$		$-$		$+$
$\frac{x-5}{3(x-2)}$	$+$	$\bullet$	$-$		$+$

$x \in (-\infty, 2) \cup [5, +\infty)$ — Final answer.

■

3.26

$$\frac{x+1}{x-7} \geq 2 + \frac{1}{x-7}$$

$\frac{x+1}{x-7} - \frac{2x-14}{x-7} - \frac{1}{x-7} \geq 0$

$\frac{-x+14}{x-7} \geq 0$

$\frac{x-14}{x-7} \leq 0$

$x \neq 7 \quad N_C \in \{7, 14\}$ Final answer.

Create a table of signs.

		7		14	
$x-7$	$-$		$+$		$+$
$x-14$	$-$		$-$		$+$
$\frac{x-14}{x-7}$	$+$	$\bullet$	$-$		$+$

$x \in (7, 14]$ Final answer.

■

3.27

$$\frac{x^2}{x+1} \geq -5 - \frac{1}{x+1}$$

$\frac{x^2}{x+1} + \frac{5x+5}{x+1} + \frac{1}{x+1} \geq 0$

$\frac{x^2+5x+6}{x+1} \geq 0$

$\frac{(x+3)(x+2)}{x+1} \geq 0$ — Factor by grouping.

$x \neq -1 \quad N_C \in \{-3, -2, -1\}$

Create a table of signs.

	-3	-2	-1	
$x+3$	$-$	$+$	$+$	$+$
$x+2$	$-$	$-$	$+$	$+$
$x+1$	$-$	$-$	$-$	$+$
$\frac{(x+3)(x+2)}{x+1}$	$-$	$+$	$-$	$\bullet$ $+$

$x \in [-3, -2] \cup (-1, +\infty)$ — Final answer.

■

3.28

$$\frac{x}{(2-x)(x+1)} \geq 1$$

$\frac{x}{(2-x)(x+1)} - \frac{(2-x)(x+1)}{(2-x)(x+1)} \geq 0$

$\frac{x}{(2-x)(x+1)} - \frac{-x^2+x+2}{(2-x)(x+1)} \geq 0$

$\frac{x^2-2}{(2-x)(x+1)} \geq 0$

$\frac{(x-\sqrt{2})(x+\sqrt{2})}{(2-x)(x+1)} \geq 0$ — Factor using difference of two squares.

$x \notin \{-1, 2\}$ $\quad N_C \in \{-\sqrt{2}, -1, \sqrt{2}, 2\}$

Create a table of signs.

		$-\sqrt{2}$		-1		$\sqrt{2}$		2	
$x + \sqrt{2}$	$-$		$+$		$+$		$+$		$+$
$x + 1$	$-$		$-$		$+$		$+$		$+$
$x - \sqrt{2}$	$-$		$-$		$-$		$+$		$+$
$2 - x$	$+$		$+$		$+$		$+$		$-$
$\frac{(x-\sqrt{2})(x+\sqrt{2})}{(2-x)(x+1)}$	$-$		$+$	$\bullet$	$-$		$+$	$\bullet$	$-$

$x \in [-\sqrt{2}, -1) \cup [\sqrt{2}, 2)$ — Final answer.

■

3.29

$$\frac{2x-7}{x^2-6x+8} \leq 1$$

$\frac{2x-7}{x^2-6x+8} - \frac{x^2-6x+8}{x^2-6x+8} \leq 0$

$\frac{-x^2+8x-15}{x^2-6x+8} \leq 0$

$\frac{x^2-8x+15}{x^2-6x+8} \geq 0$

$\frac{(x-5)(x-3)}{(x-4)(x-2)} \geq 0$ Factor by grouping.

$x \notin \{2,4\}$ $N_C \in \{2,3,4,5\}$

Create a table of signs.

		2		3		4		5	
$x-2$	$-$		$+$		$+$		$+$		$+$
$x-3$	$-$		$-$		$+$		$+$		$+$
$x-4$	$-$		$-$		$-$		$+$		$+$
$x-5$	$-$		$-$		$-$		$-$		$+$
$\frac{(x-5)(x-3)}{(x-4)(x-2)}$	$+$	$\bullet$	$-$		$+$	$\bullet$	$-$		$+$

$x \in (-\infty, 2) \cup [3,4) \cup [5, +\infty)$ Final answer.

■

3.30

$$\frac{x}{x+1} \geq \frac{2}{x+3}$$

$\frac{x}{x+1} - \frac{2}{x+3} \geq 0$

$\frac{x^2+3x}{(x+1)(x+3)} - \frac{2x+2}{(x+1)(x+3)} \geq 0$ — lcm $= (x+1)(x+3)$

$\frac{x^2+x-2}{(x+1)(x+3)} \geq 0$

$\frac{(x+2)(x-1)}{(x+1)(x+3)} \geq 0$ — Factor by grouping.

$x \notin \{-3, -1\}$ $\quad N_C \in \{-3, -2, -1, 1\}$

Create a table of signs.

	-3	-2	-1	1	
$x + 3$	$-$	$+$	$+$	$+$	$+$
$x + 2$	$-$	$-$	$+$	$+$	$+$
$x + 1$	$-$	$-$	$-$	$+$	$+$
$x - 1$	$-$	$-$	$-$	$-$	$+$
$\frac{(x+2)(x-1)}{(x+1)(x+3)}$	$+$	$-$	$+$	$-$	$+$

$x \in (-\infty, -3) \cup [-2, -1) \cup [1, +\infty)$ — Final answer.

■

3.31

$$\frac{4}{2x-1} < \frac{1}{x+1}$$

$\frac{4}{2x-1} - \frac{1}{x+1} < 0$

$\frac{4x+4}{(2x-1)(x+1)} - \frac{2x-1}{(2x-1)(x+1)} < 0$ lcm $= (2x-1)(x+1)$

$\frac{2x+5}{(2x-1)(x+1)} < 0$

$x \notin \{-1, \frac{1}{2}\}$ $N_C \in \{-\frac{5}{2}, -1, \frac{1}{2}\}$

Create a table of signs.

	$-\frac{5}{2}$	-1	$\frac{1}{2}$	
$2x + 5$	$-$	$+$	$+$	$+$
$x + 1$	$-$	$-$	$+$	$+$
$2x - 1$	$-$	$-$	$-$	$+$
$\frac{2x+5}{(2x-1)(x+1)}$	$-$	$+$	$-$	$+$

$x \in (-\infty, -\frac{5}{2}) \cup (-1, \frac{1}{2})$ Final answer.

■

3.32

$$\frac{6}{3w-4} \geq w+1$$

$\frac{6}{3w-4} - \frac{(w+1)(3w-4)}{3w-4} \geq 0$

$\frac{6}{3w-4} - \frac{3w^2-w-4}{3w-4} \geq 0$

$\frac{-3w^2+w+10}{3w-4} \geq 0$

$\frac{3w^2-w-10}{3w-4} \leq 0$

$\frac{3w^2-6w+5w-10}{3w-4} \leq 0$ Factor by grouping.

$\frac{(3w+5)(w-2)}{3w-4} \leq 0$

$x \neq \frac{4}{3}$ $N_C \in \{-\frac{5}{3}, \frac{4}{3}, 2\}$

Create a table of signs.

	$-\frac{5}{3}$	$\frac{4}{3}$		2
$3w+5$	$-$	$+$	$+$	$+$
$3w-4$	$-$	$-$	$+$	$+$
$w-2$	$-$	$-$	$-$	$+$
$\frac{(3w+5)(w-2)}{3w-4}$	$-$	$+$	$-$	$+$

$x \in (-\infty, -\frac{5}{3}] \cup (\frac{4}{3}, 2]$ Final answer.

■

3.33

$$\frac{2x+1}{4} \leq \frac{2x}{3} + \frac{1}{6}$$

$12(\frac{2x+1}{4}) \leq 12(\frac{2x}{3} + \frac{1}{6})$	lcm $= 12$
$6x + 3 \leq 8x + 2$	
$-2x \leq -1$	
$x \geq \frac{1}{2}$	
$x \in [\frac{1}{2}, +\infty)$	Final answer.
	■

3.34

$$|x-2| \geq 3$$

$x-2 \geq 3$ $x \geq 5$	Solve positive case.
$x-2 \leq -3$ $x \leq -1$	Solve negative case.
$x \in (-\infty, -1] \cup [5, +\infty)$ ■	Final answer. Combine intervals.

3.35

$$|3-2x| > 2$$

$3-2x > 2$ $-2x > -1$ $x < \frac{1}{2}$	Solve positive case.
$3-2x < -2$ $-2x < -5$ $x > \frac{5}{2}$	Solve negative case.
$x \in (-\infty, \frac{1}{2}) \cup (\frac{5}{2}, +\infty)$	Final answer. Combine intervals. ■

3.36

$$|3 + 2x| + 8 \leq 7$$

$\lvert 3 + 2x \rvert \leq -1$	
$x \in \emptyset$	Final answer. The absolute value of a real number cannot be negative. ■

3.37

$$|3x-1|+2 \geq 1$$

$\lvert 3x-1\rvert \geq -1$	
$3x-1 \geq -1$ $3x \geq 0$ $x \geq 0$	Solve positive case.
$3x-1 \leq 1$ $3x \leq 2$ $x \leq \frac{2}{3}$	Solve negative case.
$x \in \mathbb{R}$ ■	Final answer. Combine intervals.

3.38

$$\left|x^2 - 6x + 7\right| \geq 2$$

$x^2 - 6x + 7 \geq 2$ — Solve positive case.

$x^2 - 6x + 5 \geq 0$

$(x-5)(x-1) \geq 0$ — Factor by grouping.

$N_C \in \{1, 5\}$

Create a table of signs.

		1	5
$x - 1$	$-$	$+$	$+$
$x - 5$	$-$	$-$	$+$
$(x-1)(x-5)$	$+$	$-$	$+$

$x \in (-\infty, 1] \cup [5, +\infty)$

$x^2 - 6x + 7 \leq -2$ — Solve negative case.

$x^2 - 6x + 9 \leq 0$

$(x-3)^2 \leq 0$ — Factor by grouping.

$x = 3$ — The square of a real number cannot be negative.

$x \in (-\infty, 1] \cup \{3\} \cup [5, +\infty)$ — Final answer. Combine intervals.

■

3.39

$$\left|\frac{3x+9}{2x-2}\right| < -3$$

$x \in \emptyset$	Final answer. The absolute value of a real number cannot be negative.

■

3.40

$$\left|\frac{4}{x}+1\right| \geq 3$$

$\frac{4}{x}+1 \geq 3$ — Solve positive case.

$\frac{4}{x}-2 \geq 0$

$\frac{4}{x}-\frac{2x}{x} \geq 0$

$\frac{4-2x}{x} \geq 0$

$\frac{x-2}{x} \leq 0$

$x \neq 0 \quad N_C \in \{0,2\}$

Create a table of signs.

		0		2	
x	$-$		$+$		$+$
$x-2$	$-$		$-$		$+$
$\frac{x-2}{x}$	$+$	$\bullet$	$-$		$+$

$x \in (0,2]$

$\frac{4}{x}+1 \leq -3$ — Solve negative case.

$\frac{4}{x}+4 \leq 0$

$\frac{4}{x}+\frac{4x}{4} \leq 0$

$\frac{4x+4}{x} \leq 0$

$\frac{x+1}{x} \leq 0$

$x \neq 0 \quad N_C \in \{-1,0\}$

Create a table of signs.

		-1		0	
$x+1$	$-$		$+$		$+$
x	$-$		$-$		$+$
$\frac{x+1}{x}$	$+$		$-$	$\bullet$	$+$

$x \in [-1,0)$

$x \in [-1,0) \cup (0,2]$ — Final answer. Combine intervals.

■

3.41

$$\left|\frac{x}{2x-3}\right| \leq 1$$

Solve positive case.

$\frac{x}{2x-3} \leq 1$

$\frac{x}{2x-3} - \frac{2x-3}{2x-3} \leq 0$

$\frac{-x+3}{2x-3} \leq 0$

$\frac{x-3}{2x-3} \geq 0$

$x \neq \frac{3}{2}$ $\quad N_C \in \{\frac{3}{2}, 3\}$

Create a table of signs.

		$\frac{3}{2}$		3	
$2x - 3$	$-$		$+$		$+$
$x - 3$	$-$		$-$		$+$
$\frac{x-3}{2x-3}$	$+$	$\bullet$	$-$		$+$

$x \in (-\infty, \frac{3}{2}) \cup [3, +\infty)$

Solve negative case.

$\frac{x}{2x-3} \geq -1$

$\frac{x}{2x-3} + \frac{2x-3}{2x-3} \geq 0$

$\frac{3x-3}{2x-3} \geq 0$

$\frac{x-1}{2x-3} \geq 0$

$x \neq \frac{3}{2}$ $\quad N_C \in \{1, \frac{3}{2}\}$

Create a table of signs.

		1		$\frac{3}{2}$	
$x - 1$	$-$		$+$		$+$
$2x - 3$	$-$		$-$		$+$
$\frac{x-1}{2x-3}$	$+$		$-$	$\bullet$	$+$

$x \in (-\infty, 1] \cup (\frac{3}{2}, +\infty)$

$x \in (-\infty, 1] \cup [3, +\infty)$

Final answer. Intersect intervals.

■

3.42

$$0 < |x - 5| < 2$$

$\lvert x-5\rvert > 0$	Solve left inequality.
$x - 5 > 0$	Solve positive case.
$x > 5$	
$x - 5 < 0$	Solve negative case.
$x < 5$	
$x \in (-\infty, 5) \cup (5, +\infty)$	Combine intervals.
$\lvert x-5\rvert < 2$	Solve right inequality.
$x - 5 < 2$	Solve positive case.
$x < 7$	
$x - 5 > -2$	Solve negative case.
$x > 3$	
$x \in (3, 7)$	Intersect intervals.
$x \in (3, 5) \cup (5, 7)$	Final answer. Intersect intervals.
	■

3.43

$$3|8-x|+2<7-2|x-8|$$

$x<8$	$3(8-x)+2<7-2(8-x)$	Solve case where $x<8$.
$x<8$	$24-3x+2<7-16+2x$	
$x<8$	$5x>35$	
$x<8$	$x>7$	
	$x\in(7,8)$	
$x\geq 8$	$3(x-8)+2<7-2(x-8)$	Solve case where $x\geq 8$.
$x\geq 8$	$3x-24+2<7-2x+16$	
$x\geq 8$	$5x<45$	
$x\geq 8$	$x<9$	
	$x\in[8,9)$	
	$x\in(7,9)$	Final answer. Combine intervals.
		■

3.44

$$\left|\frac{9-2x}{4x}\right| \geq 1$$

$\frac{9-2x}{4x} \geq 1$ — Solve positive case.

$\frac{9-2x}{4x} - \frac{4x}{4x} \geq 0$

$\frac{9-6x}{4x} \geq 0$

$\frac{3-2x}{4x} \geq 0$

$x \neq 0 \quad N_C \in \{0, \frac{3}{2}\}$

Create a table of signs.

		0		$\frac{3}{2}$	
$4x$	$-$		$+$		$+$
$3 - 2x$	$+$		$+$		$-$
$\frac{3-2x}{4x}$	$-$	$\bullet$	$+$		$-$

$x \in (0, \frac{3}{2}]$

$\frac{9-2x}{4x} \leq -1$ — Solve negative case.

$\frac{9-2x}{4x} + \frac{4x}{4x} \leq 0$

$\frac{2x+9}{4x} \leq 0$

$x \neq 0 \quad N_C \in \{-\frac{9}{2}, 0\}$

Create a table of signs.

		$-\frac{9}{2}$		0	
$2x + 9$	$-$		$+$		$+$
$4x$	$-$		$-$		$+$
$\frac{2x+9}{4x}$	$+$		$-$	$\bullet$	$+$

$x \in [-\frac{9}{2}, 0)$

$x \in [-\frac{9}{2}, 0) \cup (0, \frac{3}{2}]$ — Final answer.

■

4

Solve the following word problems algebraically.

4.1

Find 3 consecutive even numbers whose sum is 78.

$x + (x + 2) + (x + 4) = 78$ $3x + 6 = 78$ $3x = 72$ $x = 24$	
The 3 consecutive even numbers are 24, 26, and 28.	Final answer. ■

4.2

If the length of each side of a square is increased by 2 units, the resulting square has an area that is 16 units more than the area of the original square. Determine the dimensions of the original square.

$(x+2)^2 = x^2 + 16$	The original square has a side length of x.
$x^2 + 4x + 4 = x^2 + 16$	
$4x = 12$	
$x = 3$	
The original square has a side length of 3 units.	Final answer.
	■

4.3

Susan is 9 years older than Ivy. Four years ago, the sum of their ages was 17 years. In how many years will Susan turn 18?

$(I-4)+((I+9)-4)=17$	
$2I+1=17$	
$2I=16$	
$I=8$	
$S=17$	$S=I+9$
Susan will turn 18 in 1 year.	Final answer.
	■

4.4

Ray and Joyce leave a park at the same time in separate directions. Ray drives 6 kilometeres home while Joyce drives 7 kilometers home. They arrive at the same time at their respective homes. If Ray drives 3 kilometers per hour slower than Joyce, what are their average rates?

$\frac{6}{r-3} = \frac{7}{r}$

$r(r-3)(\frac{6}{r-3}) = r(r-3)(\frac{7}{r})$ lcm $= r(r-3)$.

$6r = 7r - 21$

$r_J = 21$

$r_R = 18$

Ray drives at a speed of 18 km/h, while Joyce drives at a speed of 21 km/h.

■

4.5

To exclusively rent a videoke room, a certain number of people have to pay equal amounts to a total of Php 720.00. If there were 3 persons less, each person would then have to contribute Php 40.00 more. How many people were there originally?

$\frac{720}{x} + 40 = \frac{720}{x-3}$	
$x(x-3)(\frac{720}{x} + 40) = x(x-3)(\frac{720}{x-3})$	lcm $= x(x-3)$
$720x - 2160 + 40x(x-3) = 720x$	
$\cancel{720x} - 2160 + 40x^2 - 120x = \cancel{720x}$	
$40x^2 - 120x - 2160 = 0$	
$x^2 - 3x - 54 = 0$	
$(x-9)(x+6) = 0$	Factor by grouping.
$x \in \{-6, 9\}$	
$x = 9$	The number of people cannot be negative.
9 people were there there originally.	Final answer.

■

4.6

The numerator of a positive fraction is 5 more than thrice its denominator. If 13 is added to the denominator and 9 is subtracted from the numerator, the resulting fraction is twice the reciprocal of the original number. Find the original fraction.

$2(\frac{x}{3x+5}) = \frac{3x-4}{x+13}$	
$\frac{2x}{3x+5} = \frac{3x-4}{x+13}$	
$(3x+5)(x+13)(\frac{2x}{3x+5}) = (3x+5)(x+13)(\frac{3x-4}{x+13})$	lcm $= (3x+5)(x+13)$
$2x(x+13) = (3x+5)(3x-4)$	
$2x^2 + 26x = 9x^2 + 3x - 20$	
$7x^2 - 23x - 20 = 0$	
$x = \frac{-(-23)\pm\sqrt{(-23)^2-4(7)(-20)}}{2(7)}$	Use the quadratic formula.
$x = \frac{23\pm\sqrt{529+560}}{14}$	
$x = \frac{23\pm\sqrt{1089}}{14}$	
$x = \frac{23\pm\sqrt{9}\sqrt{121}}{14}$	
$x = \frac{23\pm 33}{14}$	
$x \in \{\frac{23+33}{14}, \frac{23-33}{14}\}$	
$x \in \{\frac{56}{14}, -\frac{10}{14}\}$	
$x \in \{4, -\frac{5}{7}\}$	
$x = 4$	We want a positive fraction.
The original fraction is $\frac{17}{4}$.	Final answer.

■

4.7

Two friends living 39 miles apart leave their homes at the same time on bicycles and travel towards each other. If one person averages 2 miles per hour more then the other and they meet in $\frac{3}{2}$ hours, what is each person's average cycling speed?

$\frac{3}{2}x + \frac{3}{2}(x+2) = 39$ $3x + 3x + 6 = 78$ $6x = 72$ $x = 12$
The slower person has a speed of 12 mph, while the faster person has a speed of 14 mph. Final answer. ■

4.8

Bus A leaves a terminal at 12:00 noon while Bus B headed for the same destination leaves at the same terminal 15 minutes later. After covering 75 kilometers, Bus B catches up to Bus A at 1:15 pm. Determine the speed of each bus.

$t_A = 75\,\text{min} = \frac{5}{4}\text{h}$ $\frac{5}{4}x = 75$ $\frac{5x}{4} = 75$ $5x = 300$ $x = 60$	
Bus A travels at 60 km/h, while Bus B travels at 75 km/h.	Final answer.
	■

4.9

One alloy contains 80% gold (by weight) and another contains 55% gold. How many grams of each alloy must be combined to make 40 grams of an alloy that is 70% gold?

$0.8x + 0.55(40 - x) = 0.7(40)$ $0.8x + 22 - 0.55x = 28$ $0.25x - 6$ $x = 24$	
24 g and 16 g of the 80% and 55% gold alloys, respectively, are needed. ■	Final answer.

4.10

One acid soluton is 60% water and another acid solution is 40% water. How many liters of each solution should be mixed to produce 90 liters of acid solution which is 50% water?

$0.6x + 0.4(90 - x) = 0.52(90)$ $0.6x + 36 - 0.4x = 46.8$ $0.2x = 10.8$ $x = 54$	
54 L and 36 L of the acid solutions with 60% and 40% water, respectively, are needed. ■	Final answer.

4.11

A grocer wishes to mix two types of coffee beans. One costs Php 40.00 per kilogram and the other costs Php 60.00 per kilogram. How much of each type must be mixed to obtain 50 kilograms of coffee that reasonably costs Php 55.00 per kilogram?

$40x + 60(50 - x) = 55(50)$ $40x + 3000 - 60x = 2750$ $-20x = -250$ $x = 12.5$	
12.5 kg and 37.5 kg of the Php 60.00 and Php 55.00 per kilo coffee, respectively, are needed.	Final answer.

■

4.12

Wes invested a total of PhP $120,000.00$ in two accounts. One account earns 7% annual interest while the other earns 5% annual interest. If a year after the investment, Wes earns Php $7,400.00$, how much did he put in each account?

$0.07x + 0.05(120000 - x) = 7400$ $0.07x + 6000 - 0.05x = 7400$ $0.02x = 1400$ $x = 70000$	
Php $70,000.00$ and Php $50,000.00$ are invested in the 7% and 5% annual interest accounts, respectively.	Final answer.

■

4.13

If a rowing team rows with the current, they can cover 16 miles in 2 hours. Against the current, they can cover 10 miles in 4 hours. Find their rate of rowing in calm water and the rate of the current.

$\frac{16}{2} - \frac{10}{4} = (r + c) - (r - c)$ $\frac{32}{4} - \frac{10}{4} = r + c - r + c$ $\frac{22}{4} = 2c$ $c = \frac{11}{4}$	Find the speed of the current.
$r + \frac{11}{4} = \frac{16}{2}$ $r = \frac{32}{4} - \frac{11}{4}$ $r = \frac{21}{4}$	Find the speed of the rowing team in calm water.
The rowing team has a speed of $\frac{21}{4}$mph in calm water, while the current has a speed of $\frac{11}{4}$mph. ■	Final answer.

4.14

Mary can clean an office in 10 hours. Working together, John and Mary can clean the office in 7 hours. How long would it take John to clean the office by himself?

$\frac{1}{10} + \frac{1}{x} = \frac{1}{7}$	
$70x(\frac{1}{10} + \frac{1}{x}) = 70x(\frac{1}{7})$	lcm $= 70x$
$7x + 70 = 10x$	
$3x = 70$	
$x = \frac{70}{3}$	
John will take $\frac{70}{3}$h to clean the office alone.	Final answer.
	■

4.15

An office has two photocopy machines. Machine A can photocopy 1000 pieces of a single file in 5 hours, while Machine B can photocopy 1000 pieces of the same file in 4 hours. How long will it take the two machines, both working together to photocopy 1600 pieces of a file?

$\frac{1000}{5} + \frac{1000}{4} = \frac{1600}{x}$	
$20x(\frac{1000}{5} + \frac{1000}{4}) = 20x(\frac{1600}{x})$	lcm $= 20x$
$4000x + 5000x = 32000$	
$9000x = 32000$	
$x = \frac{32000}{9000}$	
$x = \frac{32}{9}$	
Both machines working together will take $\frac{32}{9}$h to make 1600 copies of a file.	Final answer.

■

4.16

A man can paint a room by himself in 8 hours while his son can paint the same room in 12 hours. The man works by himself for 3 hours and is then joined by his son. How many hours will it take them to finish painting?

$u = 1 - 3\frac{1}{8}$ $u = \frac{5}{8}$	Find the unpainted amount of room.
$\frac{1}{8} + \frac{1}{12} = \frac{5}{8x}$ $24x(\frac{1}{8} + \frac{1}{12}) = 24x(\frac{5}{8x})$ $3x + 2x = 15$ $5x = 15$ $x = 3$	lcm $= 24x$
Both of them will take 3 h to paint the still-unpainted part of the room. ■	Final answer.

4.17

A woman can do a certain job in 8 hours, while her daughter can do it in 12 hours. After the woman and the daughter worked for an hour, they were joined by the woman's son, and the three completed the job in three hours. How long would it take the son to finish the same job alone?

$u = 1 - \frac{1}{8} - \frac{1}{12}$ $u = \frac{24}{24} - \frac{3}{24} - \frac{2}{24}$ $u = \frac{19}{24}$	Find the uncompleted part of the job.
$\frac{1}{8} + \frac{1}{12} + \frac{1}{x} = \frac{19}{24(3)}$ $\frac{1}{8} + \frac{1}{12} + \frac{1}{x} = \frac{19}{72}$ $72x(\frac{1}{8} + \frac{1}{12} + \frac{1}{x}) = 72x(\frac{19}{72})$ $9x + 6x + 72 = 19x$ $15x + 72 = 19x$ $4x = 72$ $x = 18$	lcm $= 72x$
Her son will take 18 h to do the job alone.	Final answer.

■